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Focus on AEDs

Prevalence of Automated External Defibrillators at Cardiac Arrest High-risk Sites

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Abstract

Objective. Automated external defibrillators (AEDs) distributed throughout communities may improve survival from cardiac arrest. The purpose of this study was to determine if AEDs were present at high-risk locations for cardiac arrest in King County, Washington. **Methods.** The authors compiled a list of sites based on a five-year study that identified public sites with the highest incidence of cardiac arrests in King County. They conducted a structured telephone survey with the manager, director, or owner of those high-risk sites. **Results.** Of the 263 identified high-risk cardiac arrest sites, we obtained information for 228 (87%) sites. Overall, 87 of 228 (38%) high-risk sites had one or more AEDs. The AED dissemination varied greatly by type of site. The airport, the two county jails, the five public sports venues, and the nine ferries/train terminals each reported at least one AED on site. In contrast, none of the 13 shelters and 19% of health clubs/gyms reported an AED on site. Nearly half (44%) of sites without AEDs cited cost as a factor preventing them from purchasing AEDs in the future. **Conclusion.** Although AEDs have diffused into high-risk sites in this community, the diffusion appears to vary by the type of site. **Key words:** sudden cardiac arrest; emergency medical services; defibrillation.

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Prompt defibrillation after collapse from ventricular fibrillation cardiac arrest increases the likelihood of survival. Automated external defibrillators (AEDs) provide the opportunity to improve survival from ventricular fibrillation cardiac arrest because they enable laypersons not trained in rhythm recognition to deliver potentially lifesaving therapy. One approach to the distribution of AEDs is to place them in public locations throughout the community that are at high risk of cardiac arrest. Although high-risk public locations have been identified,[1–3] little is known about the actual dissemination of AEDs into these locations. The purpose of this investigation was to survey locations previously identified as high risk for cardiac arrest in King County, Washington, to determine if AEDs were present and, if not, to determine the reasons for not instituting an AED program.

Methods

Study Design and Setting

The study was a prospective survey study, conducted from June 16, 2003, through July 12, 2003, of all sites in King County, Washington, that were defined as high risk for cardiac arrest. King County, including the city of Seattle, has a population of 1.7 million (2000 census). The median age of residents in the county is 35.7 years, with a per-capita income of $29,521. Whites comprise 78.9% of the population, Asian Americans 12.5%, and blacks or African Americans 6.5%. The University of Washington Human Subjects Committee approved the study.

Selection of Survey Sites

Survey sites were selected based on results of a prior study that identified public sites with the highest risk of cardiac arrests in public locations in King County.[1] The high-risk sites had an annual incidence of cardiac arrest ranging from 7.00 to 0.03. These sites included the international airport, county jails, large shopping malls, public sports venues, large industrial sites, golf courses, shelters, ferry/train terminals, health clubs/gyms, and community/senior centers. Under the heading "health

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clubs/gyms," we did not include swimming pools (unless the pool was part of a health club), aerobics studios, Pilates studios, yoga studios, or rowing centers, because they were not included in the original study of high-risk cardiac arrest sites. We compiled a list of addresses and phone numbers for sites falling into the above categories using Qwestdex.com, an online Yellow Pages database. Those sites that were listed in the database but whose phone number has been disconnected (n = 15) were dropped from the study and presumed no longer in business. In addition, the Department of Public Health of King County has maintained a registry of AEDs since 1999. From this database we abstracted information about the number and location of registered AEDs at sites that we were unable to contact. This occurred in four instances, for one large shopping mall, one large industrial site, and two health clubs/gyms.

Data Collection

We conducted a structured telephone survey in which the manager, director, or owner of each site was asked:

1. Have you heard of an automated external defibrillator (AED) and in what kind of situation an AED might be used?
2. How many AEDs are currently available at your facility?
3. (If no AEDs are present:) Are you planning to purchase an AED(s) in the future? If you do not have an AED and are not planning to purchase an AED, what is the primary reason?
4. (If AEDs are present:) How many employees currently work at your facility? How many are currently trained to use an AED?

We attempted to contact each site up to four times, at various times and days. In assessing training at public sports venues, we used the number of trained personnel for non-game days. (Game days increased the number of paramedics and emergency medical technicians [EMTs] on standby at the stadium.)

Statistical Analysis

Descriptive statistics were used to assess the proportion of high-risk sites that had an AED on site, the number of persons trained, and the potential reasons for not instituting an AED program. Results were stratified by the type of site according to the original classification of high-risk sites.[1] The analyses were performed with SPSS statistical software (version 11.5; SPSS, Inc., Chicago, IL).

Results

Of the 263 eligible high-risk cardiac arrest sites in King County that were identified, 224 surveys were completed and an additional four sites were registered in the Public Health AED registry. Thus, information was available for 228 of 263 sites (87%). Of the 35 sites without information, eight declined to participate in the survey and 27 could not be reached despite four contact attempts.

Overall, 87 of 228 (38%) high-risk sites had one or more AEDs. There were a total of 413 AEDs within the 87 sites (overall median = 1 [range 1 to 200]) (Table 1). The AED dissemination varied greatly by type of site. The airport, the two county jails, the five public sports venues, and the nine ferries/train terminals each reported at least one AED on site. In contrast, none of the 13 shelters and 19% of health clubs/gyms reported an AED on site.

Table 1. On-site Automated External Defibrillator (AED) Programs and Training According to High-risk Location Type

Location Category	Number of Sites	Number of Sites Successfully Contacted*	Number of Sites with AED†	Total Number of AEDs	Median Number of AEDs among Sites with AEDs (Upper and Lower Range)	Total Number of Personnel Trained	Median Number of Personnel Trained among Sites with AEDs (Lower and Upper Range)
International airport	1	1	1 (100%)	200	200	UNK‡	UNK‡
County jail	2	2	2 (100%)	3	1.5 (1–2)	110	55 (30–80)
Larger shopping mall	11	10	5 (50%)	7	1 (1–2)	142	40 (12–50)
Public sports venue	6	5	5 (100%)	51	5 (1–37)	158	30 (23–40)
Large industrial site	21	19	18 (95%)	79	2 (1–13)	576§	9 (4–145)
Golf course	31	28	11 (39%)	22	2 (1–3)	283	20 (10–70)
Shelters	16	13	0 (0%)	0	NA	NA	NA
Ferries/train terminal	10	9	9 (100%)	9	1 (1–1)	9	1 (1–1)
Health club/gym	138	114	22 (19%)	27	1 (1–3)	580‖	23 (0–150)‖
Community/senior center	27	27	14 (52%)	15	1 (1–2)	133¶	20 (0–150)¶
Total	**263**	**228**	**87 (38%)**	**413**	**1 (1–200)**	**1993**	**20 (0–150)**

*This includes the four sites that could not be contacted but were in the Health Department AED registry.
†The number of sites successfully contacted was used as the denominator to determine the percentage of sites with AEDs.
‡One site could not provide a count of the number of personnel trained. Median could not be calculated.
§One site could not provide a count of the number of personnel trained. Median based on 17 sites.
‖Six sites could not provide a count of the number of personnel trained. Median based on 16 sites.
¶Two sites could not provide a count of the number of personnel trained. Median based on 12 sites.

Of the 87 sites with AEDs, nine sites could not provide an accurate count of the number of trained personnel for their site. At the 78 sites that could provide accurate counts, a total of 1,993 personnel had been trained, with a median of 20 personnel trained per site (range 0 to 150). One site had an AED, but the person trained in its use had vacated his position; at the time of the survey, the site had not trained a new person.

Of the 141 sites without an AED, 19 reported that they had plans to purchase AEDs. Additionally, 14 sites had never heard of an AED. Of the 108 sites that were familiar with the AED but did not have an on-site AED and did not plan to purchase an AED, nearly half (48 of 108, 44%) of the high-risk locations cited cost as the major factor, and less than 10% (8 of 108, 7%) cited liability concern as the primary factor (Table 2). No site specifically stated that instituting an AED program would be ineffective.

DISCUSSION

Automated external defibrillators distributed among high-risk cardiac arrest sites in a community have the potential to reduce the number of deaths occurring each year from cardiac arrest. Preliminary findings of the national Public Access Defibrillation Trial suggest that a program of AED placement at high-risk sites improves survival from cardiac arrest.[4] (The Public Access Defibrillation trial findings were presented at a "Breaking News" presentation of the American Heart Association and have not yet been published.) We found that 38% of sites previously identified as high risk in our community have instituted an AED program. However, AED dissemination into high-risk sites varied considerably, depending on the type of high-risk location. Among those without an AED on site, the cost of the AED was most often cited as the reason for not instituting an AED program; liability concerns were not commonly reported.

Over the last several years, many factors may have in some manner promoted the concept of public-access defibrillation. The considerable public health toll of out-of-hospital cardiac arrest and the generally poor survival have highlighted a need for innovative approaches to improve outcomes from cardiac arrest. High-risk public sites for cardiac arrest have been identified. Several publications in peer-reviewed scientific journals have reported encouraging results with layperson first-responder use of AEDs in particular settings.[5–7] Conversely, the results of a randomized trial evaluating an approach to public-access defibrillation have not yet been published. Local and national programs have encouraged and promoted public-access AED programs. In some instances, AED placement has been mandated by law.[8] Media have increasingly reported generally positive experiences of AED use. A Lexis–Nexis search (academic version) for the year 1998 revealed that "automated external defibrillator" was mentioned in 50 articles (42 times in major newspapers and eight times in magazines). In 2002, "automated external defibrillator" appeared in 149 articles (141 times in major newspapers and eight times in magazines). Technical and interface advances have improved the AED, and manufacturers have extensively marketed the device. The 38% AED penetration into our community's high-risk public sites should be considered in this context. It is likely that particular factors along with personal preference were differentially important in making the decision to institute an AED program.

The dissemination of AEDs into high-risk sites varied considerably depending on the type of site, indirectly supporting the notion that particular factors differentially influence a site's decision to institute an AED program. Large industrial businesses, sporting venues, transportation centers, and jails almost uniformly were equipped with AEDs. Golf courses and community/senior centers had intermediate AED penetration. Health clubs/gyms and shelters were uncommonly equipped with AEDs. Health clubs/gyms and shelters also generally employed fewer persons and were smaller businesses. These characteristics would

TABLE 2. Reasons Cited for Not Instituting an Automated External Defibrillator AED Program

Location Category	Cost	Proximity to Fire Department or Hospital	Liability	No Need for AED, Site Risk Too Low/Other	Familiar with AED, but Have Not Considered/Contemplated On-site AED Program	Unfamiliar with an AED
Larger shopping mall	1	0	1	0	0	0
Large industrial site	0	0	0	1	0	0
Golf course	11	0	1	3	2	0
Shelters	6	0	0	1	4	2
Ferries/train terminal	0	0	0	0	0	0
Health club/gym	23	10	5	9	18	12
Community/senior center	7	2	1	0	2	0
Total	**48**	**12**	**8**	**14**	**26**	**14**

potentially contribute to decreased awareness of the AED and economic barriers to AED purchase, reasons typically cited for not instituting an AED program.

Among sites with an AED, all reported some form of initial formal training. However, the number of personnel trained per AED varied considerably both across the different types of high-risk sites and within some of the site types. This variability may, to some extent, reflect differences in operational hours, staffing, or the physical design of the facility. However, an alternative interpretation is that too many or too few persons are being trained. If too few are trained, an optimal response to an on-site cardiac arrest may be jeopardized. Conversely, training too many persons may not be the best use of resources for a particular site.

We believe communities could benefit from surveys similar to that reported here. Our survey identified three types of sites with low penetration, namely, golf courses, shelters, and health clubs. A community could use this type of data to target AED placements.

Limitations

The study has limitations. This investigation was a survey, so it is possible, though we believe unlikely, that sites could have incorrectly or falsely reported the presence or absence of an AED. When searching for sites we used Qwestdex.com; although it is not designed specifically for the study, it yielded a more up-to-date list of sites than the general Yellow Pages. The dissemination of AEDs was assessed only for predefined high-risk sites; the extent to which AEDs have been incorporated in other settings was not determined. Although King County is a large metropolitan community with a heterogeneous population, the results may not be generalizable to other communities. For example, there appear to be strong community awareness and training for cardiac arrest in King County: two AED manufacturers are headquartered there, and 11 high-risk sites had AEDs as part of a federally funded public-access defibrillation trial (known as the PAD trial). These factors would likely increase the dissemination of AEDs into high-risk cardiac arrest sites.

Conclusion

In our community, 38% of high-risk public sites have purchased an AED and trained personnel, though the extent appears to vary substantially depending on the type of site. Cost considerations were the most common reason reported for not instituting an AED program. Improving survival in cardiac arrest will require a multifaceted approach. Placement of AEDs in public high-risk sites constitutes one method to incrementally address the challenge. The results of this study suggest that this strategy has been partially implemented.

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